UNIT-3
Multiple encryptions and triple DES

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Outline

- Multiple encryption and triple DES
- Electronic Code Book Mode
- Cipher Block Chaining Mode
- Cipher Feedback Mode
- Output Feedback Mode
- Counter Mode

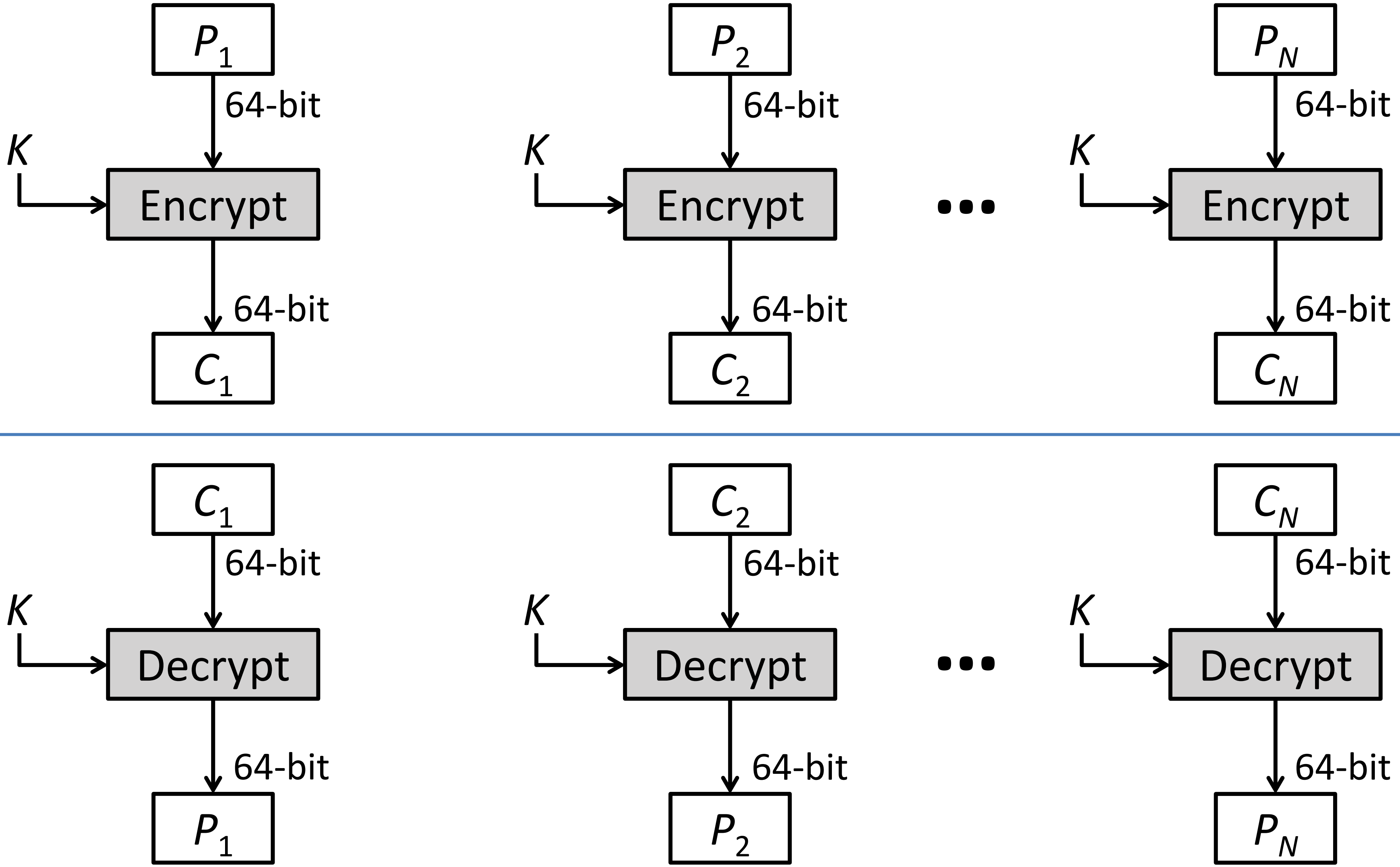
Block Cipher Modes of Operations

- To apply a block cipher in a **variety of applications**, five "modes of operation" have been defined.
- The **five modes** are intended to cover a wide variety of applications of encryption for which a block cipher could be used.
- These modes are intended for use with any symmetric block cipher, including triple DES and AES.
  1. Electronic Code Book (ECB)
  2. Cipher Block Chaining (CBC)
  3. Cipher Feedback (CFB)
  4. Output Feedback (OFB)
  5. Counter (CTR)

1. Electronic Code Book (ECB)

- In **ECB** Mode Plaintext handled one block at a time and each block of plaintext is encrypted using the same key.
- The term **codebook** is used because, for a given key, there is a unique ciphertext for every b-bit block of plaintext.

1. ECB Encryption & Decryption



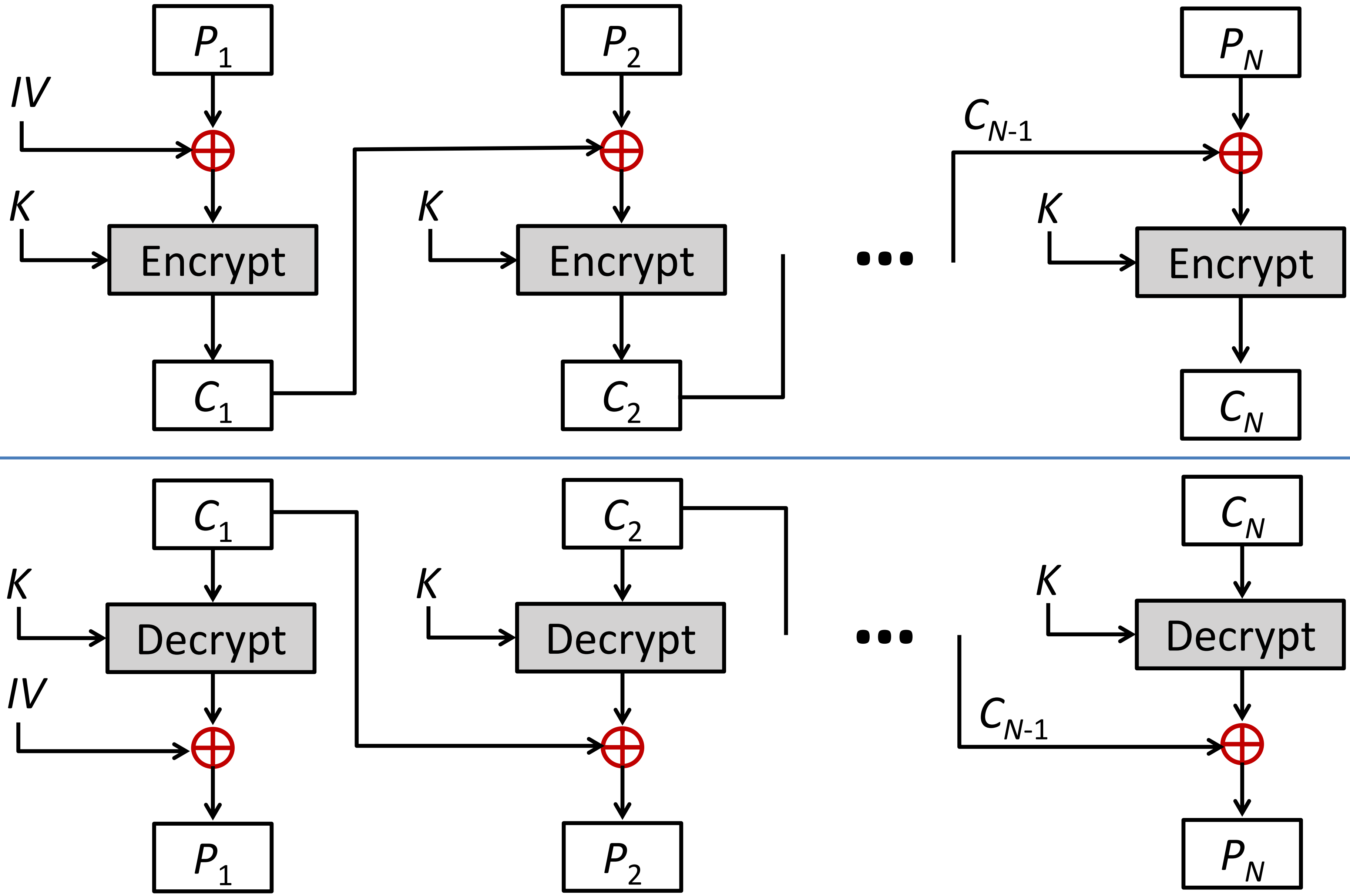
Electronic Code Book - Cont...

- **Strength:** it's simple.
- **Weakness:**
  - Repetitive information contained in the plaintext may show in the ciphertext, if aligned with blocks.
  - If the message has repetitive elements with a period of repetition a multiple of b bits, then these elements can be identified by the analyst.
- **Typical application:**
  - Secure transmission of short pieces of information (e.g. a temporary encryption key)

2. Cipher Block Chaining (CBC)

- **CBC** is a technique in which the same plaintext block, if repeated, produces different ciphertext blocks.
- In this scheme, the input to the encryption algorithm is the **XOR** of the current plaintext block and the preceding ciphertext block; the same key is used for each block.
- To produce the first block of ciphertext, an **initialization vector (IV)** is XORed with the first block of plaintext.
- On decryption, the **IV** is XORed with the output of the decryption algorithm to recover the first block of plaintext.

2. CBC - Encryption & Decryption



2. Cipher Block Chaining (CBC) – Cont…

- **Strength:** because of the chaining mechanism of CBC, it is an appropriate mode for encrypting messages of length greater than b bits
- **Typical application:**
  - General-purpose block oriented transmission
  - Authentication

CBC	$C_1 = \mathrm{E}(K, [P_1 \oplus \mathrm{IV}])$ $C_j = \mathrm{E}(K, [P_j \oplus C_{j-1}])\ j = 2, \ldots, N$	$P_1 = \mathrm{D}(K, C_1) \oplus \mathrm{IV}$ $P_j = \mathrm{D}(K, C_j) \oplus C_{j-1}\ j = 2, \ldots, N$

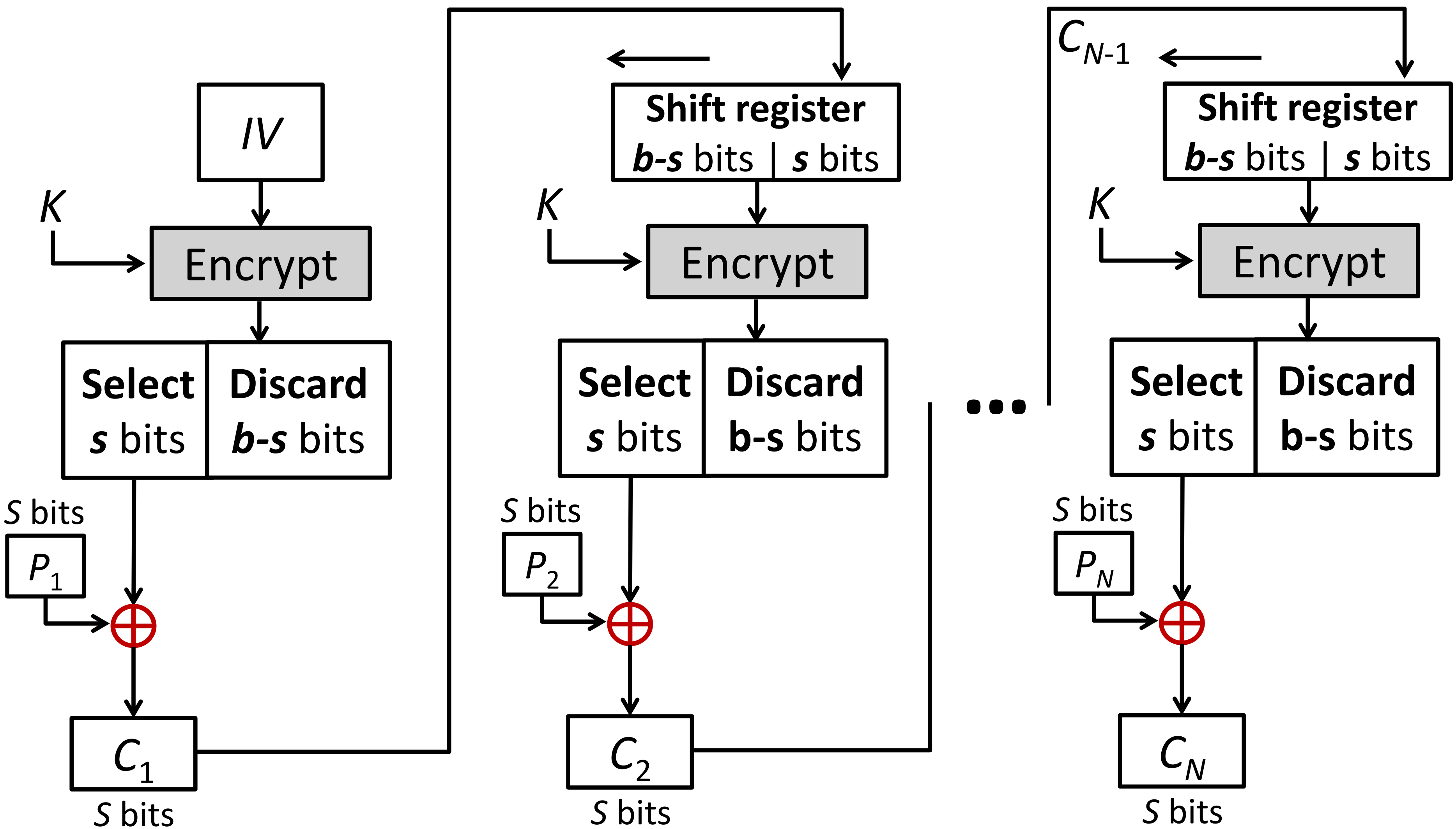
3. Cipher Feedback Mode (CFB)

- For AES, DES, or any block cipher, encryption is performed on a block of b bits. In DES, b = 64 and in AES, b = 128.

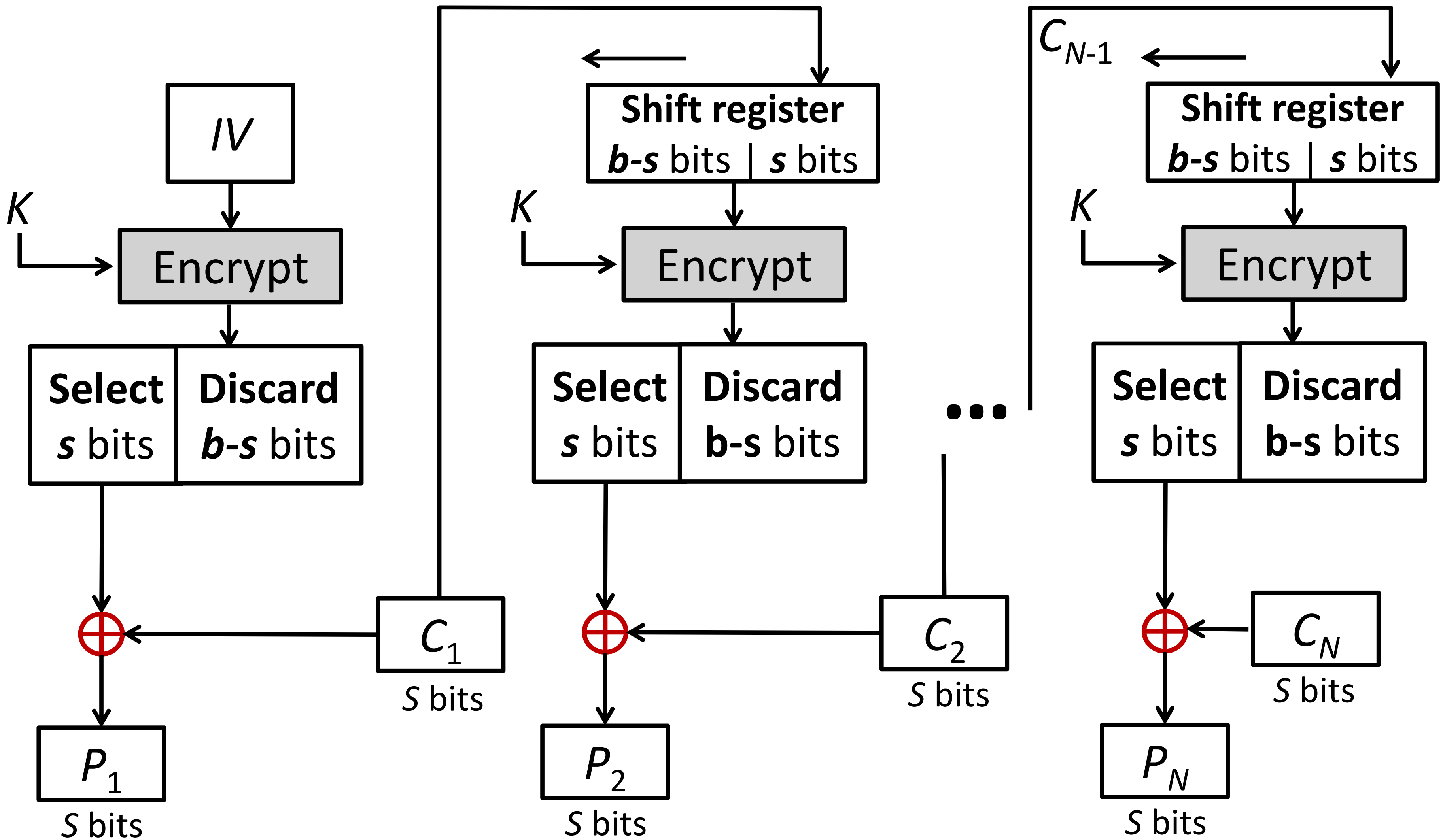
- However, it is possible to convert a block cipher into a stream cipher, using cipher feedback (CFB) mode, output feedback (OFB) mode, and counter (CTR) mode.

- A stream cipher eliminates the need to pad a message to be an integral number of blocks.

3. CFB Encryption



3. CFB Decryption



CFB Mode

- The input to the encryption function is a **b-bit shift register** that is initially set to some initialization vector (IV).
- The leftmost (most significant) **s bits** of the output of the encryption function are XORed with the first segment of plaintext **P1** to produce the first unit of ciphertext **C1** , which is then transmitted.
- In addition, the contents of the shift register are shifted left by **s bits**, and C1 is placed in the rightmost (least significant) s bits of the shift register.
- For decryption, the same scheme is used, except that the received ciphertext unit is XORed with the output of the encryption function to produce the plaintext unit.

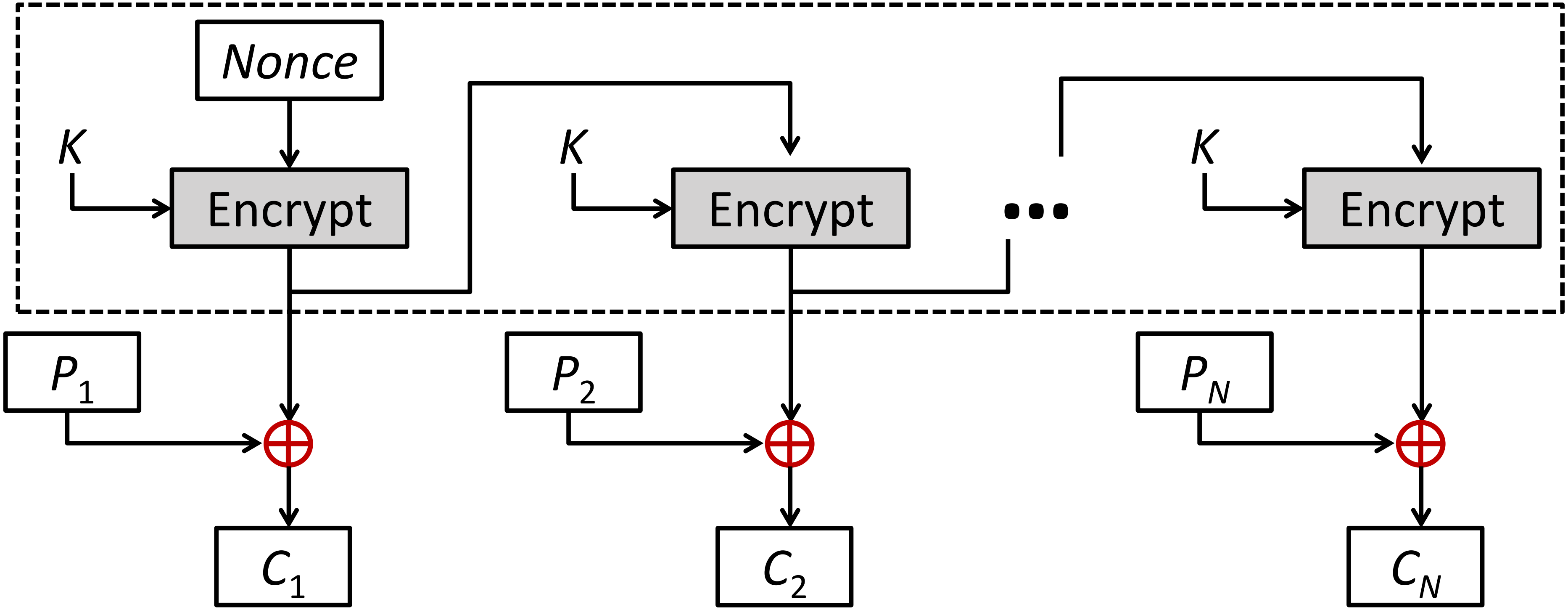
CFB Mode – Cont…

CFB	$I_1 = IV$ $I_j = \mathrm{LSB}_{b-s}(I_{j-1}) \parallel C_{j-1} \quad j = 2, \ldots, N$ $O_j = \mathrm{E}(K, I_j) \quad j = 1, \ldots, N$ $C_j = P_j \oplus \mathrm{MSB}_s(O_j) \quad j = 1, \ldots, N$	$I_1 = IV$ $I_j = \mathrm{LSB}_{b-s}(I_{j-1}) \parallel C_{j-1} \quad j = 2, \ldots, N$ $O_j = \mathrm{E}(K, I_j) \quad j = 1, \ldots, N$ $P_j = C_j \oplus \mathrm{MSB}_s(O_j) \quad j = 1, \ldots, N$

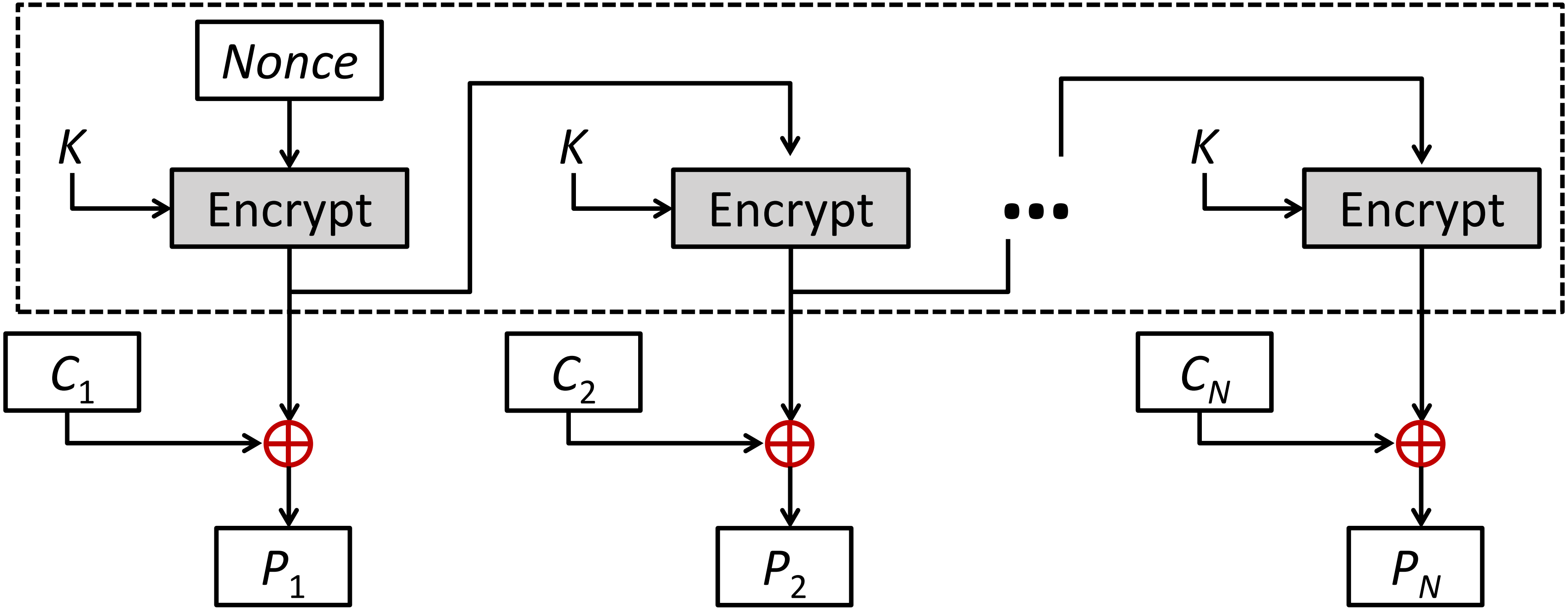
4. Output Feedback Mode (OFB)

- The output feedback (**OFB**) mode is similar in structure to that of **CFB**.
- For **OFB**, the output of the encryption function is fed back to become the input for encrypting the next block of plaintext.
- In **CFB**, the output of the XOR unit is fed back to become input for encrypting the next block.
- The other difference is that the **OFB** mode operates on full blocks of plaintext and ciphertext, whereas **CFB** operates on an **s-bit** subset.
- **Nonce:** A time-varying value that has at most a negligible chance of repeating, for example, a random value that is generated anew for each use, a timestamp, a sequence number, or some combination of these.

4. OFB Encryption



4. OFB Decryption



OFB Mode

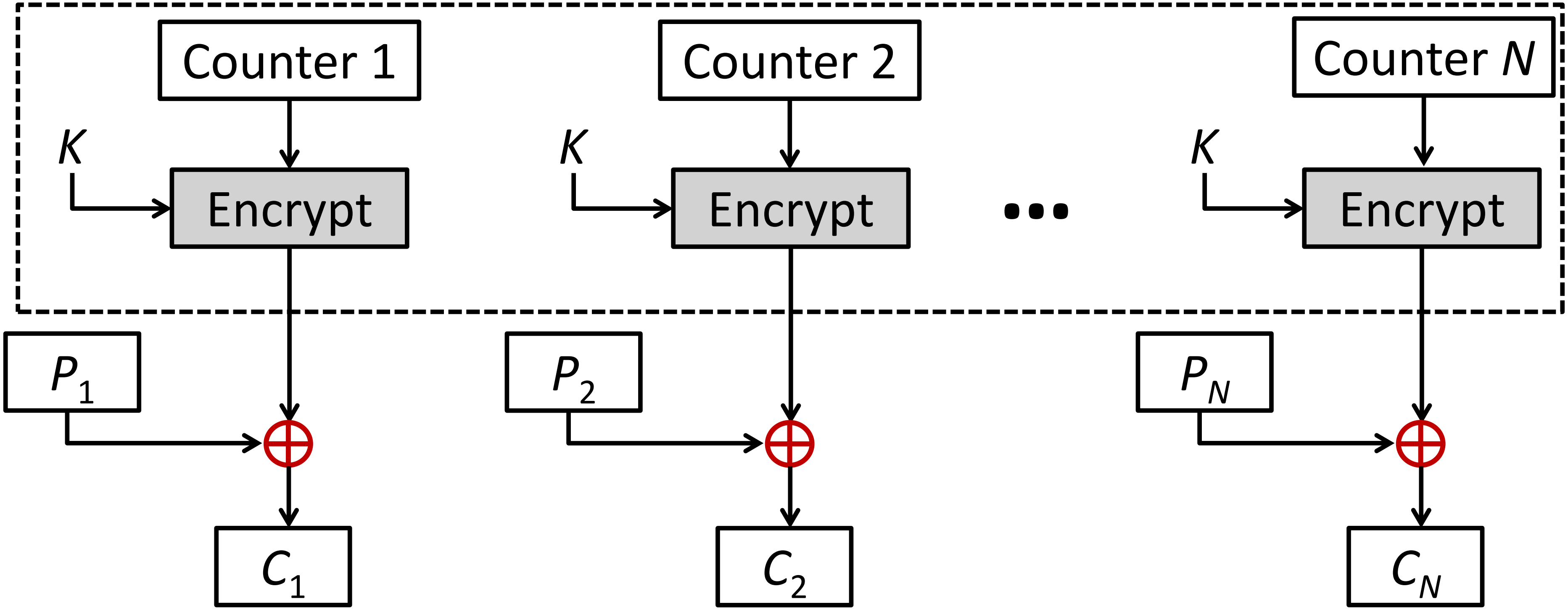
- Each bit in the ciphertext is independent of the previous bit or bits.
- This avoids error propagation
- Pre-compute of forward cipher is possible

OFB	$I_1 = Nonce$ $I_j = O_{j-1} \quad j = 2, \ldots, N$ $O_j = \mathrm{E}(K, I_j) \quad j = 1, \ldots, N$ $C_j = P_j \oplus O_j \quad j = 1, \ldots, N-1$ $C_N^* = P_N^* \oplus \mathrm{MSB}_u(O_N)$	$I_1 = Nonce$ $I_j = O_{j-1} \quad j = 2, \ldots, N$ $O_j = \mathrm{E}(K, I_j) \quad j = 1, \ldots, N$ $P_j = C_j \oplus O_j \quad j = 1, \ldots, N-1$ $P_N^* = C_N^* \oplus \mathrm{MSB}_u(O_N)$

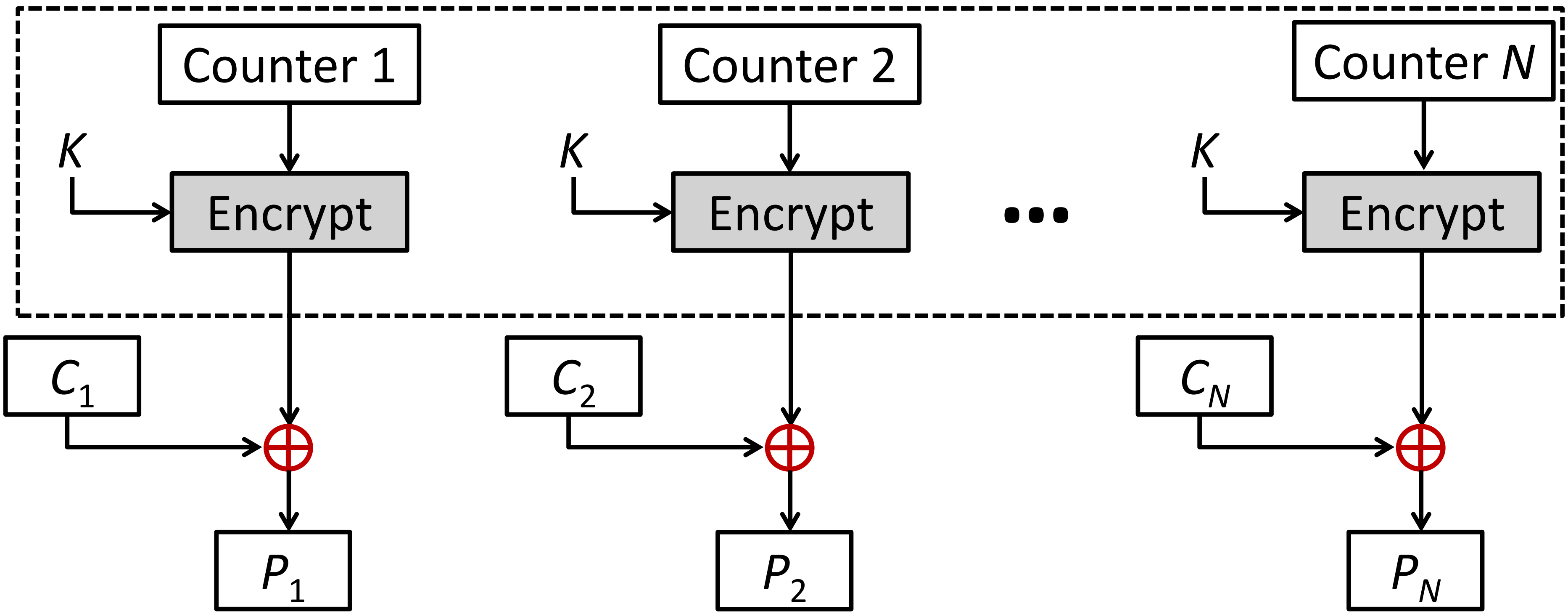
5. Counter Mode (CTR)

- Counter (**CTR**) mode has increased recently with applications to ATM (asynchronous transfer mode) network security and IP sec (IP security).
- A counter equal to the plaintext block size is used.
- The counter value must be different for each plaintext block that is encrypted.
- Typically, the counter is initialized to some value and then incremented by 1 for each subsequent block

5. CTR Encryption



4. CTR Decryption



CTR	$C_j = P_j \oplus \mathrm{E}(K, T_j) \qquad j = 1, \ldots, N-1$ $C_N^* = P_N^* \oplus \mathrm{MSB}_u[\mathrm{E}(K, T_N)]$	$P_j = C_j \oplus \mathrm{E}(K, T_j) \qquad j = 1, \ldots, N-1$ $P_N^* = C_N^* \oplus \mathrm{MSB}_u[\mathrm{E}(K, T_N)]$

Advantages of the CTR Mode

- Strengths:
  - Needs only the encryption algorithm
  - Random access to encrypted data blocks
  - blocks can be processed (encrypted or decrypted) in parallel
  - Simple; fast encryption/decryption

- Counter must be
  - Must be unknown and unpredictable
  - pseudo-randomness in the key stream is a goal

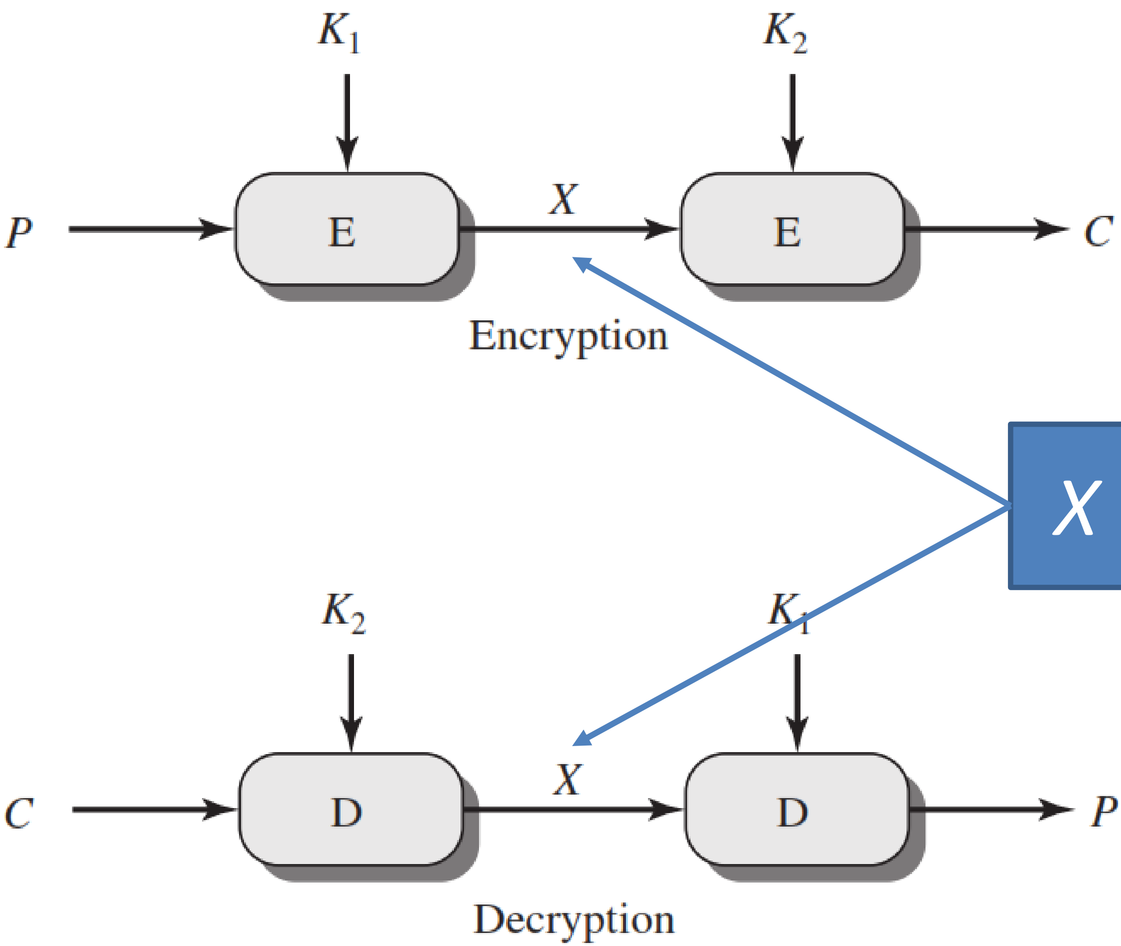
Summary of all modes

Operation Mode	Description	Type of Result
ECB	Each n-bit block is encrypted independently with same key	Block Cipher
CBC	Same as ECB, but each block is XORed with previous cipher text	Block Cipher
CFB	Each s-bit block is XORed with s-bit key which is part of previous cipher text	Stream Cipher
OFB	Same as CFB, but the shift register is updated by the previous s-bit key	Stream Cipher
CTR	Same as OFB, but a counter is used instead of nonce	Stream Cipher

Multiple Encryption

- Given the potential vulnerability of **DES** to a brute-force attack, there has been considerable interest in finding an alternative.
- One approach is to design a completely new algorithm, of which **AES** is a prime example.
- Another alternative, which would preserve the existing investment in software and equipment, is to use **multiple encryption with DES and multiple keys**.

Double DES



$C = \mathrm{E}(K_2, \mathrm{E}(K_1, P))$

$X = \mathrm{E}(K_1, P) = \mathrm{D}(K_2, C)$

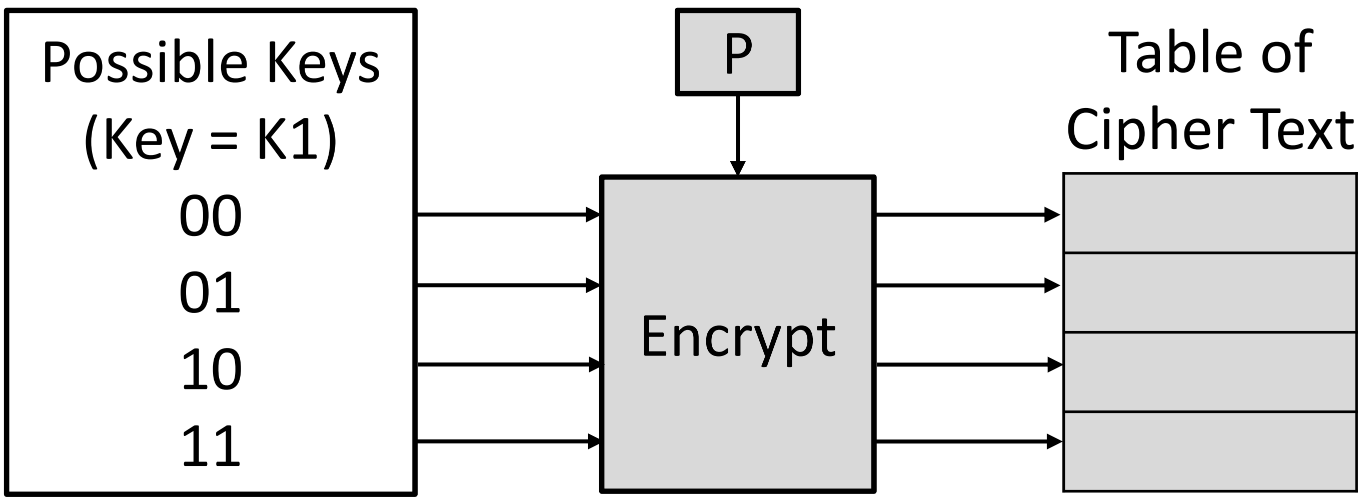
$P = \mathrm{D}(K_1, \mathrm{D}(K_2, C))$

Meet in the Middle Attack

- This attack involves encryption from one end, decryption from the other and matching the results in the middle.
- Suppose cryptanalyst knows **P** and corresponding **C**.
- Now, the aim is to obtain the values of K_1 and K_2.

Meet in the Middle Attack Step-1

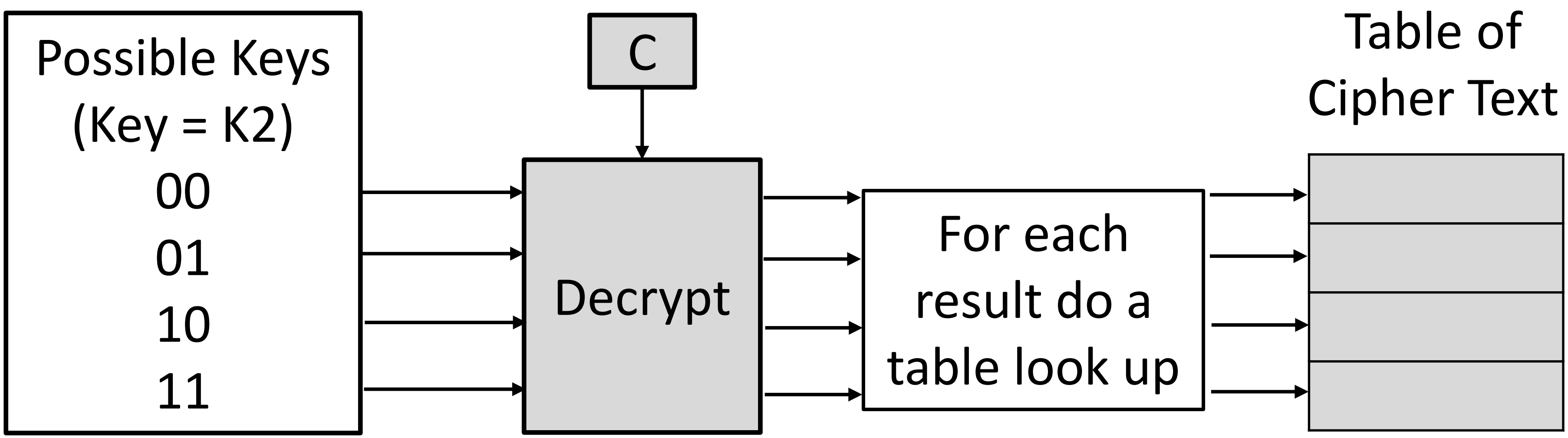
- For all possible values (2^{56}) of key K1, the cryptanalyst would encrypt the P by performing E(K1,P).
- The cryptanalyst would store output in a table.



Cryptanalyst encryption operation

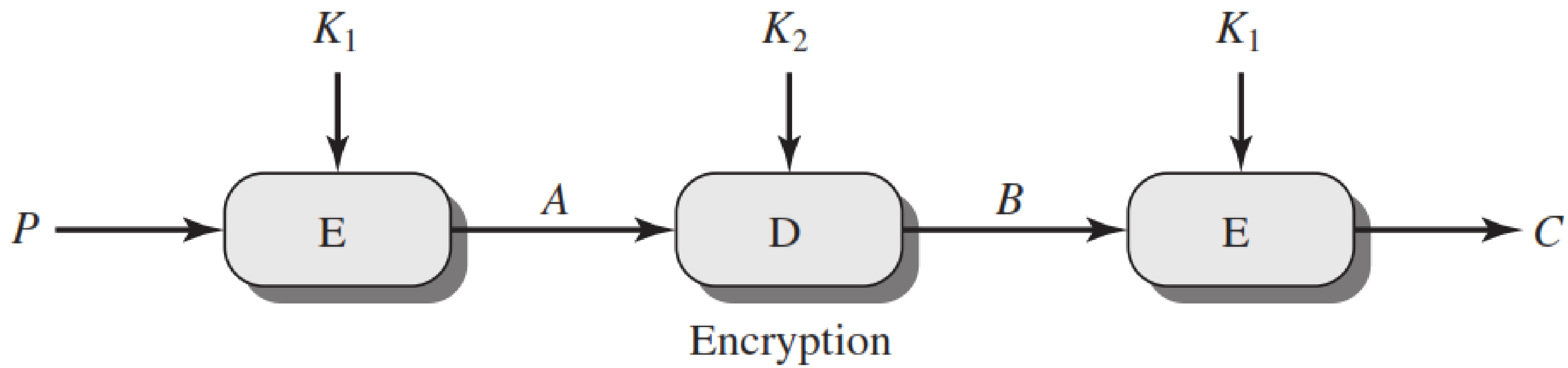
Meet in the Middle Attack Step-2

- Cryptanalyst decrypt the known **C** with all possible values of **K2**.
- In each case cryptanalyst will **compare** the resulting value with the all values in the table of ciphertext.

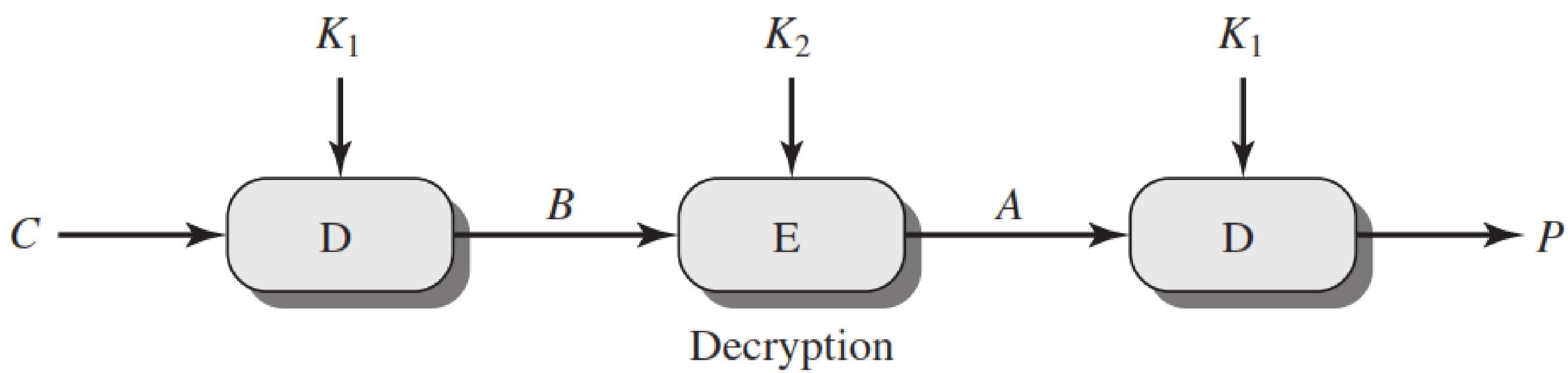


Cryptanalyst decryption operation

Triple DES



C=E(K_1,D(K_2, E(K_1,P)))



P=D(K_1,E(K_2, D(K_1,C)))

THANK YOU FOR

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